

MOBAN Chapter 7 — Part 2: LTE

Comprehensive Study Guide

PART 1: THEORY SUMMARY

1. Introduction

1.1 Motivation and Context

LTE (Long Term Evolution) is a 4G cellular standard developed by 3GPP. It was motivated by several converging pressures on the cellular industry:

Driver	Detail
Higher data rates	3G (UMTS/HSPA) data rates were becoming insufficient for mobile broadband demand
Quality of Service	End-to-end QoS guarantees for diverse traffic types (voice, video, data)
Packet-switched optimization	Eliminate circuit-switched overhead; all services over IP
Cost reduction	Flatter architecture reduces network element count and operational cost
Low complexity	Simpler protocol stack to ease implementation and reduce latency

LTE satisfies the ITU **beyond IMT-2000** requirements:

- Peak downlink rate of **100 Mbps**
- Fully all-IP network architecture

1.2 LTE Requirements

Requirement	Target
Downlink peak rate	100 Mbps (at 20 MHz bandwidth)
Uplink peak rate	50 Mbps (at 20 MHz bandwidth)
Active users per cell	200 users per cell at 5 MHz bandwidth
Mobility	Up to 350 km/h (at reduced performance)
RAN round-trip time	< 10 ms (radio access network only)
Spectrum flexibility	1.4, 3, 5, 10, 15, 20 MHz channel bandwidths
MIMO	2×2 and 4×4 antenna configurations

1.3 Duplex Modes

LTE supports both duplex methods:

Mode	Name	Description
FDD	Frequency Division Duplex	Separate frequencies for uplink and downlink; can operate simultaneously
TDD	Time Division Duplex	Same frequency; uplink and downlink separated in time

Belgium spectrum allocation example:

Operator	Mode
Proximus	FDD
BASE / Telenet	FDD
Orange	FDD
BUCD / Dense Air	TDD

1.4 Scale

LTE had **4.69 billion subscriptions** globally in Q3 2025, making it the dominant mobile broadband technology worldwide.

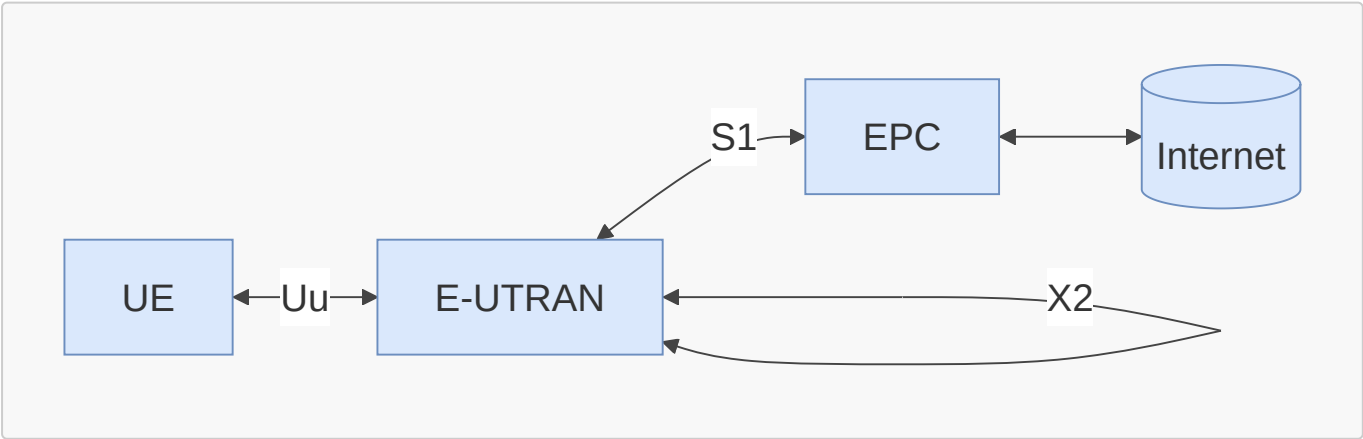
2. Architecture

2.1 Overall Structure

The LTE system is formally called the **EPS (Evolved Packet System)**:

EPS = UE + E-UTRAN + EPC

Diagram — LTE End-to-End Architecture



Architecture element explanations:

Abbreviation	Full Name	Contains	Role
UE	User Equipment	Client device (phone, modem)	User terminal
E-UTRAN	Evolved UMTS Terrestrial Radio Access Network	eNodeBs	Radio access, cell-level mobility, Radio Resource Management
EPC	Evolved Packet Core	MME, SGW, PGW, HSS	Core switching, mobility management, gateway to internet
Internet	—	—	External packet data network

Interfaces at a glance:

Interface	Between	Purpose
Uu	UE ↔ eNodeB	Radio interface
S1	eNodeB ↔ EPC	Core network access
X2	eNodeB ↔ eNodeB	Direct inter-eNodeB signaling (handover)

2.2 Flat Architecture vs. UMTS

LTE introduced a "flat architecture" — fewer network hierarchy levels compared to UMTS:

Aspect	UMTS	LTE
RAN elements	NodeB + RNC (two levels)	eNodeB only (one level)
Core elements	SGSN + GGSN (two levels)	SGW + PGW
Intelligence in RAN	RNC held most RRM logic	Each eNodeB is fully capable
Round-trip delays	Longer (more hops)	Shorter (fewer hops)

Key benefit: by removing the RNC layer and pushing RRM intelligence into the eNodeB, LTE achieves shorter round-trip delays and lower latency — critical for the < 10 ms RAN RTT requirement.

2.3 Evolved Packet Core (EPC) Components

MME (Mobility Management Entity)

- User profile management
- Authentication and authorization of UEs
- Mobility signaling (tracking area updates)
- Tracking and paging of idle UEs
- Bearer management (setting up and releasing QoS bearers)
- Equivalent to the SGSN signaling plane in UMTS

SGW (Serving Gateway)

- Routes user-plane IP traffic between the eNodeB and the PGW

- User-plane anchor for intra-LTE handovers (inter-eNodeB) and inter-3GPP technology handovers
- Buffers downlink packets when the UE is in idle mode

PGW (Packet Data Network Gateway)

- IP address and IP prefix allocation for UEs
- Policy enforcement and charging
- Gateway to external packet data networks (internet, IMS)
- Deep packet inspection for QoS and billing

HSS (Home Subscriber Server)

- Permanent subscriber database — stores subscriber profiles, authentication vectors, service entitlements
- Functional equivalent of the HLR in GSM
- Works with MME for authentication and authorization

2.4 EPC Component Summary

Component	Full Name	Role	GSM Equivalent
MME	Mobility Management Entity	Signaling, authentication, paging, mobility	MSC + VLR (control plane)
SGW	Serving Gateway	User-plane routing, handover anchor	SGSN (user plane)
PGW	Packet Data Network Gateway	IP allocation, charging, gateway	GGSN
HSS	Home Subscriber Server	Subscriber database	HLR

3. E-UTRAN

3.1 eNodeB (evolved Node B)

The eNodeB is the only type of base station in LTE. It is far more capable than the UMTS NodeB, because it absorbs all the functions that the RNC used to perform:

Function	Description
Radio Resource Management (RRM)	Channel allocation, scheduling, power control
Scheduling	Per-subframe (1 ms) assignment of resource blocks to UEs
HARQ management	Controls hybrid ARQ retransmission
Handover decisions	Decides when and to which cell to hand over
Header compression	IP header compression for efficiency
Security	Encryption and integrity protection of the air interface

Each eNodeB:

- Can operate in **FDD mode**, **TDD mode**, or support both
- Connects to the EPC via the **S1 interface**
- Connects directly to neighboring eNodeBs via the **X2 interface**

3.2 X2 Interface

The X2 interface allows **direct eNodeB-to-eNodeB communication** without going through the core network. Key uses:

- Coordinating handover between adjacent eNodeBs
- Exchanging interference coordination information
- Forwarding buffered user-plane packets to the target eNodeB during handover

In UMTS, all inter-cell coordination had to go through the RNC. The X2 interface in LTE enables faster, more efficient handover.

4. Radio Interface

4.1 OFDM Downlink

LTE uses **OFDM (Orthogonal Frequency Division Multiplexing)** on the downlink. Key properties:

Property	Benefit
Orthogonal subcarriers	Eliminates inter-carrier interference
Long symbol duration	Robust against frequency-selective multipath fading
Cyclic prefix	Absorbs inter-symbol interference from multipath
Flexible bandwidth	Vary the number of subcarriers to match available spectrum
Broadcast/multicast	Easy to share the same signal to multiple receivers

Subcarrier spacing: 15 kHz (fixed, regardless of system bandwidth)

Bandwidth options and corresponding subcarrier counts:

Channel BW	Number of Resource Blocks	Number of subcarriers
1.4 MHz	6	72
3 MHz	15	180
5 MHz	25	300
10 MHz	50	600
15 MHz	75	900
20 MHz	100	1200

4.2 Uplink: SC-FDMA

The uplink uses **SC-FDMA (Single Carrier Frequency Division Multiple Access)** rather than OFDM. Reason: SC-FDMA has a lower **Peak-to-Average Power Ratio (PAPR)**, which extends UE battery life and allows cheaper power amplifiers in the handset.

4.3 OFDMA Multiple Access

OFDMA (Orthogonal Frequency Division Multiple Access) allows the eNodeB to schedule different subcarriers to different UEs within the same 1 ms subframe.

Resource hierarchy:

Unit	Definition	Size
Resource Element (RE)	1 subcarrier × 1 OFDM symbol	Smallest unit
Resource Block (RB)	12 subcarriers × 1 slot (0.5 ms)	84 resource elements (12 × 7)
Subframe	2 slots	1 ms scheduling unit
Radio Frame	10 subframes	10 ms

The eNodeB scheduler assigns resource blocks to UEs on a **per-subframe (1 ms)** basis, enabling fine-grained frequency and time-selective scheduling.

4.4 Frame Structure

```

Radio Frame (10 ms)
├── 10 × Subframe (1 ms each)
│   ├── 2 × Slot (0.5 ms each)
│       └── 7 OFDM symbols (normal cyclic prefix)
    
```

Parameter	Value
Radio frame duration	10 ms
Subframes per frame	10
Subframe duration	1 ms
Slots per subframe	2
Slot duration	0.5 ms
OFDM symbols per slot	7 (normal cyclic prefix)
Subcarrier spacing	15 kHz
Resource block bandwidth	12 × 15 kHz = 180 kHz
Resource elements per RB	12 subcarriers × 7 symbols = 84

4.5 Downlink Physical Channels

LTE defines several physical channels in the downlink, each with a specific role:

Channel / Signal	Full Name	Purpose
PDSCH	Physical Downlink Shared Channel	Carries user payload data
PDCCH	Physical Downlink Control Channel	Scheduling assignments and control information
PBCH	Physical Broadcast Channel	Cell parameters: bandwidth, transmit power, system frame number
PSS	Primary Synchronization Signal	Gives NID(2) — identifies the cell within its group
SSS	Secondary Synchronization Signal	Gives NID(1) — identifies the cell group
CRS	Cell Reference Signal	Enables RSRP measurement and channel estimation

RSRP (Reference Signal Received Power): the key measurement metric used for cell selection and handover decisions in LTE. Measured from the CRS.

4.6 MIMO

LTE supports **MIMO (Multiple Input Multiple Output)** antenna configurations:

Configuration	Description
2×2 MIMO	2 transmit antennas at eNodeB, 2 receive antennas at UE
4×4 MIMO	4 transmit antennas at eNodeB, 4 receive antennas at UE

MIMO enables:

- **Spatial multiplexing:** transmit independent data streams on each antenna pair — multiplies throughput
- **Beamforming:** focus radiated energy toward a specific UE — improves SNR and reduces interference
- **Transmit diversity:** same data on multiple antennas with different coding — improves reliability

5. LTE Procedures

5.1 Cell Search

Why cell search?

Before a UE can connect to a cell, it must:

1. Find a valid carrier frequency
2. Determine the cell identity
3. Synchronize its timing to the cell

Types of cell search:

Type	Context	Timing constraint
Initial cell search	UE powers on; does not know carrier frequency	No strict timing — scans all possible frequencies
Neighbor cell search	UE is connected; looking for handover target	Strict timing — must complete within measurement gaps

Cell identity structure:

LTE has **504 unique Physical Cell Identities (PCIs)**:

Parameter	Description	Range
NID(1)	Cell group identity (from SSS)	0 – 167 (168 groups)
NID(2)	Cell identity within group (from PSS)	0, 1, or 2 (3 values)
PCI	Physical Cell ID = $3 \times \text{NID}(1) + \text{NID}(2)$	0 – 503 (504 values)

Cell search sequence:

1. UE detects PSS → determines NID(2) (3 possible sequences)
2. UE detects SSS → determines NID(1) (168 possible sequences)
3. $\text{PCI} = 3 \times \text{NID}(1) + \text{NID}(2)$
4. UE decodes PBCH → obtains cell bandwidth, power, system frame number
5. UE reads system information on PDSCH → ready to attempt access

5.2 Random Access Procedure (Contention-Based)

Purpose: Establish uplink synchronization and allocate a C-RNTI (Cell Radio Network Temporary Identifier) to the UE.

Triggered by:

- Initial network access (after power-on cell search)
- Radio link failure (re-establishment)
- Handover to a new cell
- Re-entering connected state from idle

Four-step procedure:

Step	Message	Direction	Content
1	RACH Preamble	UE → eNodeB	Random preamble chosen from PRACH resource; establishes initial UL timing
2	Random Access Response (RAR)	eNodeB → UE	Timing advance correction; UL resource grant; Temporary C-RNTI (TC-RNTI)
3	RRC Connection Request	UE → eNodeB	UE identity (TMSI or random number); reason for access
4	Contention Resolution	eNodeB → UE	Echoes UE identity to resolve collisions; assigns permanent C-RNTI

Multiple UEs may choose the same preamble in step 1 (collision). The contention resolution in step 4 resolves this — only the UE that sees its own identity echoed back considers the procedure successful.

5.3 Link Adaptation (AMC and HARQ)

Goal: always use the most efficient modulation and coding scheme that the current channel quality can support, while keeping the Block Error Ratio (BLER) below **10%**.

Adaptive Modulation and Coding (AMC) flow:

```

UE measures SINR on CRS
  → UE reports CQI (Channel Quality Indicator) to eNodeB
    → eNodeB selects MCS (Modulation and Coding Scheme)
      → eNodeB schedules UE with chosen MCS

```

Modulation options:

Modulation	Bits per symbol	When used
QPSK	2	Poor channel conditions
16-QAM	4	Moderate channel conditions
64-QAM	6	Good channel conditions

FEC coding rates: range from **1/8** (very robust, low spectral efficiency) to **4/5** (high spectral efficiency, needs good SNR).

HARQ (Hybrid Automatic Repeat Request):

- When a transport block is received with errors, the UE sends a **NACK**
- The eNodeB retransmits the same data
- The UE **combines** the retransmission with the previously received (erroneous) copy — this **incremental redundancy** improves the probability of successful decoding
- Chase combining or incremental redundancy variants

TPC (Transmit Power Control):

- eNodeB sends power control commands to UEs
- Goal: minimize uplink interference while maintaining adequate SINR at the eNodeB
- UEs close to the cell center transmit at lower power; cell-edge UEs transmit at higher power

Frequency-Selective Scheduling:

- The eNodeB assigns different resource blocks to different UEs based on their instantaneous channel quality
- A UE may have a deep fade on some RBs but good quality on others
- Scheduling is done every **1 ms subframe** — exploits both **frequency diversity** and **multiuser diversity**

5.4 Handover

Hard handover only:

Unlike UMTS (which supported soft handover — simultaneous connection to multiple NodeBs), LTE uses **hard handover exclusively**:

- UE is connected to exactly one eNodeB at any moment
- Connection with the old eNodeB is released before (or immediately as) the connection with the new eNodeB is established

Types of handover:

Type	Name	Description
Horizontal	Intra-LTE handover	Handover between two LTE eNodeBs
Vertical	Inter-RAT handover	Handover between LTE and 2G or 3G

Event A3 — Handover Trigger:

The most common handover trigger in LTE. The start condition is:

Neighbor RSRP > Serving cell RSRP + Hysteresis (Hys) + Cell Individual Offset ... for a duration exceeding **TTT (Time-To-Trigger)**

The hysteresis and TTT prevent **ping-pong** — repeated handover back and forth between two cells.

Handover flow (intra-LTE, intra-MME):

1. UE measures neighbor cell RSRP (from CRS) → reports to SeNB (source eNodeB)
2. SeNB evaluates Event A3 trigger condition
3. SeNB sends Handover Request to TeNB (target eNodeB) via X2
4. TeNB admits the UE; reserves resources; sends Handover Request Acknowledge to SeNB
5. SeNB sends RRC Connection Reconfiguration to UE (includes target cell info)
6. SeNB forwards buffered DL packets to TeNB (via X2 data forwarding)
7. UE disconnects from SeNB; performs random access on TeNB
8. UE sends RRC Connection Reconfiguration Complete to TeNB

9. TeNB notifies MME; path is switched in the core network
10. TeNB releases resources at SeNB via X2

During steps 6–9, downlink traffic is **buffered at TeNB** and delivered to the UE once the connection is established — minimizing packet loss.

5.5 Voice Over LTE Options

LTE is a pure packet-switched network with no circuit-switched domain. This creates four approaches to voice calls:

Approach	Name	How it works	Quality	Complexity
OTT	Over The Top	Voice as a regular data app (Skype, WhatsApp, Signal, Viber)	Variable (best-effort)	None — no network involvement
VoLTE	Voice over LTE	Voice in PS domain using IMS; dedicated QoS bearer; AMR codec	Best — dedicated radio and core resources	High — requires IMS, QoS, coverage guarantee
CSFB	Circuit Switched Fallback	At call setup, UE is redirected to 2G/3G for a circuit-switched call	Good (legacy quality)	Medium — requires 2G/3G network and inter-RAT coordination
SRVCC	Single Radio Voice Call Continuity	Mid-call fallback from VoLTE to CS when UE leaves LTE coverage	Good (seamless transition)	High — requires coordination between LTE core and 2G/3G core

Key distinctions:

- **OTT**: no special network involvement; quality depends on data connection
- **VoLTE**: voice entirely in the packet domain with end-to-end QoS — the "proper" LTE voice solution
- **CSFB**: avoids the need for full VoLTE infrastructure by reusing 2G/3G for voice — used during early LTE deployments
- **SRVCC**: handles the edge case where a VoLTE call is active when LTE coverage ends mid-call

PART 2: SUMMARY TABLES & QUICK REFERENCE

LTE Key Parameters

Parameter	Value
Downlink multiple access	OFDMA
Uplink multiple access	SC-FDMA
Subcarrier spacing	15 kHz

Parameter	Value
Resource block bandwidth	180 kHz (12 × 15 kHz)
Resource elements per RB	84 (12 subcarriers × 7 symbols)
OFDM symbols per slot (normal CP)	7
Slot duration	0.5 ms
Subframe duration	1 ms
Radio frame duration	10 ms
Channel bandwidths	1.4, 3, 5, 10, 15, 20 MHz
Peak DL rate (20 MHz, 4×4 MIMO, 64-QAM)	~150–300 Mbps
Peak UL rate (20 MHz)	~75 Mbps
Maximum mobility	350 km/h
RAN round-trip time target	< 10 ms
Physical Cell IDs (PCIs)	504
Cell groups (NID(1))	168
Cells per group (NID(2))	3 (values 0, 1, 2)
BLER target (AMC)	< 10%
Modulation schemes	QPSK, 16-QAM, 64-QAM
FEC coding rates	1/8 to 4/5
MIMO configurations	2×2, 4×4

Architecture Components

Component	Layer	Full Name	Key Role	GSM/UMTS Equivalent
UE	Terminal	User Equipment	User terminal, radio modem	MS
eNodeB	E-UTRAN	evolved Node B	Radio access, RRM, scheduling, handover	NodeB + RNC
MME	EPC	Mobility Management Entity	Signaling, auth, paging, mobility	MSC + VLR (control)
SGW	EPC	Serving Gateway	User-plane routing, handover anchor	SGSN (user plane)
PGW	EPC	Packet Data Network Gateway	IP allocation, charging, internet gateway	GGSN

Component	Layer	Full Name	Key Role	GSM/UMTS Equivalent
HSS	EPC	Home Subscriber Server	Permanent subscriber database	HLR

Interfaces Summary

Interface	Between	Protocol / Purpose
Uu	UE ↔ eNodeB	LTE-Uu radio interface
S1-MME	eNodeB ↔ MME	Control-plane (signaling)
S1-U	eNodeB ↔ SGW	User-plane (data)
X2	eNodeB ↔ eNodeB	Handover coordination, interference management
S11	MME ↔ SGW	Mobility and bearer management
S5/S8	SGW ↔ PGW	User-plane data (S5 intra-PLMN; S8 roaming)
S6a	MME ↔ HSS	Authentication and subscription data

Physical Downlink Channels and Signals

Channel / Signal	Abbreviation	Direction	Purpose
Physical Downlink Shared Channel	PDSCH	DL	User data payload
Physical Downlink Control Channel	PDCCH	DL	Scheduling assignments, control info
Physical Broadcast Channel	PBCH	DL	Cell parameters (BW, power, SFN)
Primary Synchronization Signal	PSS	DL	NID(2) — cell identity within group
Secondary Synchronization Signal	SSS	DL	NID(1) — cell group identity
Cell Reference Signal	CRS	DL	RSRP measurement, channel estimation

Logical Channel Comparison: GSM vs. LTE

Function	GSM	LTE
User data	TCH/F, TCH/H	PDSCH (DL) / PUSCH (UL)
Control/scheduling	AGCH, PAGCH	PDCCH
Broadcast	BCCH, FCCH, SCH	PBCH, PSS, SSS, SIBs on PDSCH
Paging	PCH	PDCCH (paging message on PDSCH)

Function	GSM	LTE
Random access	RACH (slotted Aloha)	PRACH (preamble; 4-step procedure)
Measurement reference	BCCH beacon	CRS (Cell Reference Signal)

Voice Call Comparison

Method	Domain	Network requirement	When used
OTT (WhatsApp etc.)	Packet (app layer)	None (data only)	Any data connection
VoLTE	Packet (IMS)	IMS, QoS bearers, full LTE coverage	Modern LTE deployments
CSFB	Circuit-switched (2G/3G)	2G/3G network available	Early LTE; fallback
SRVCC	Circuit-switched (handover)	2G/3G + LTE + SRVCC support in core	Mid-call LTE coverage loss

Common Exam Pitfalls

1. **EPS vs. EPC:** EPS is the full system (UE + E-UTRAN + EPC); EPC is the core network only — do not use them interchangeably.
2. **LTE uses hard handover only:** unlike UMTS which supports soft handover (UE connected to multiple NodeBs simultaneously), LTE UEs are always connected to exactly one eNodeB at a time.
3. **PCI formula:** $PCI = 3 \times NID(1) + NID(2)$ — NID(1) comes from the SSS (range 0–167), NID(2) comes from the PSS (range 0–2).
4. **PSS gives NID(2), SSS gives NID(1):** the "primary" signal gives the smaller piece of information (only 3 values); the "secondary" gives the group (168 values).
5. **OFDM vs. OFDMA:** OFDM is the modulation/multiplexing technique (one user fills all subcarriers); OFDMA is the multiple-access extension (different subcarriers assigned to different users per subframe).
6. **Resource block definition:** $12 \text{ subcarriers} \times 1 \text{ slot (0.5 ms)} = 84 \text{ resource elements}$ — not $12 \times 10 \text{ ms}$ or any other combination.
7. **Scheduling granularity is 1 ms (subframe):** the eNodeB makes a new scheduling decision every subframe, not every slot or every frame.
8. **eNodeB vs. NodeB:** in UMTS, the NodeB was simple and the RNC held all intelligence; in LTE, the eNodeB is fully autonomous — it handles RRM, scheduling, HARQ, and handover decisions without a separate RNC.
9. **Flat architecture benefit is latency:** the removal of the RNC in LTE (and the reduction from SGSN+GGSN to SGW+PGW in the core) primarily reduces the number of hops and thus round-trip delays.
10. **HARQ combines — it does not simply discard and retry:** the receiver keeps erroneous packets and combines them with retransmissions for improved decoding probability. This is what makes it "hybrid" (FEC + ARQ).

11. **CQI is reported by the UE, MCS is chosen by the eNodeB:** the UE measures the channel and suggests a quality indicator; the eNodeB makes the final scheduling and modulation decision.
12. **CSFB vs. SRVCC timing:** CSFB falls back before the call is established (at call setup); SRVCC falls back during an ongoing call.
13. **Cell search sequence:** PSS first (3 NID(2) hypotheses) → SSS (168 NID(1) hypotheses) → PBCH → system information. Never the other way around.
14. **Random access C-RNTI vs. TC-RNTI:** the TC-RNTI (Temporary) is assigned in the RAR (step 2); the permanent C-RNTI is confirmed only after contention resolution (step 4).